Generative Adversial Networks (GAN)

FNN
RNN
CNN
Transformer
GAN



→ training unstable
→ hard to evaluate

⇓

Diffusion models

i) rare dataset → training
ii) Ghibli trend
iii) Deep fakes

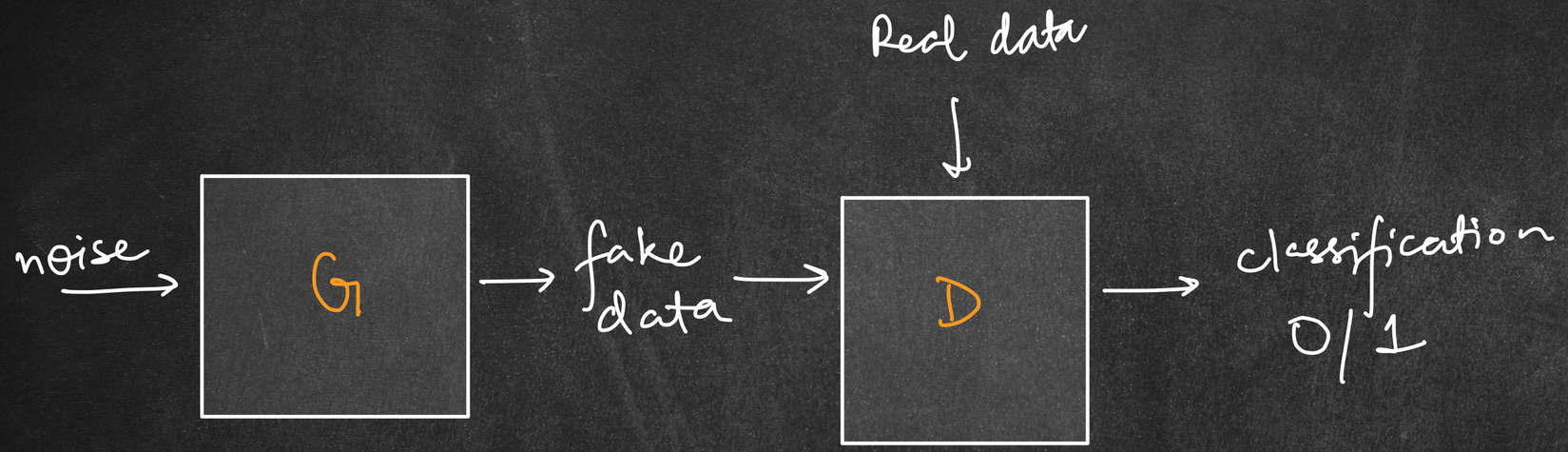
GANs

Architecture

2 NN

- Generator (G)
- Discriminator (D)

Generative Adversial Networks

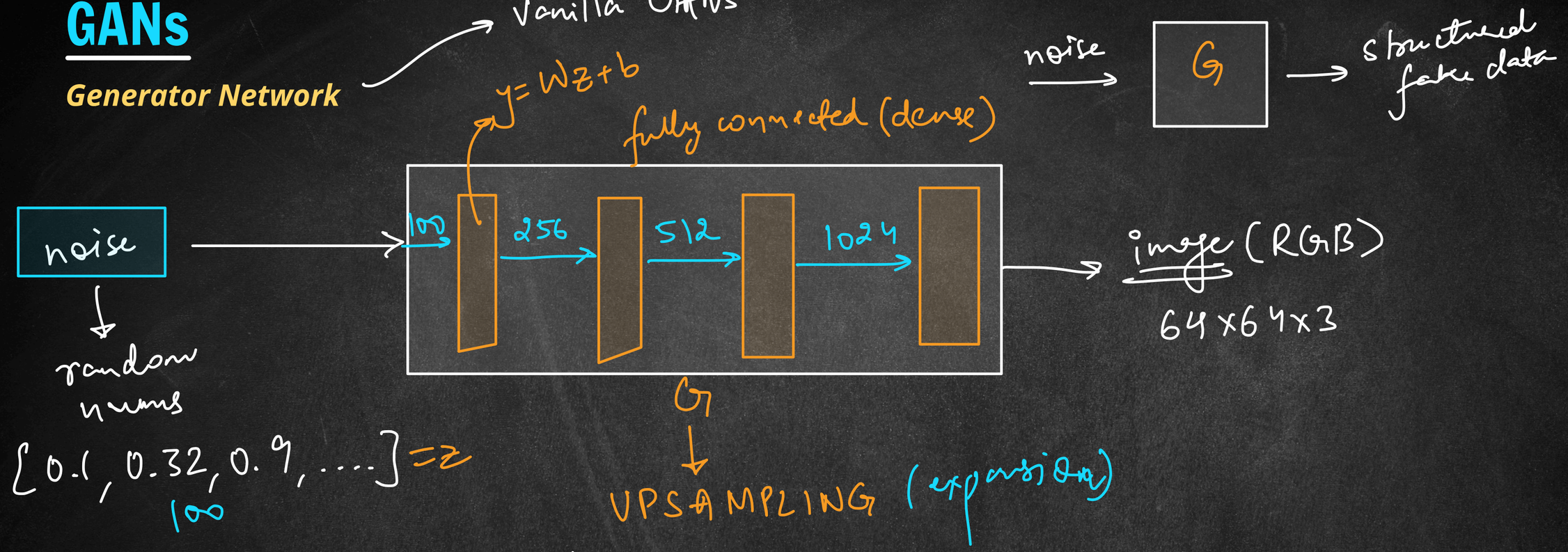


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GANs

Generator Network

Vanilla GANs



random nums

[0.1, 0.32, 0.9, ...] = z

100

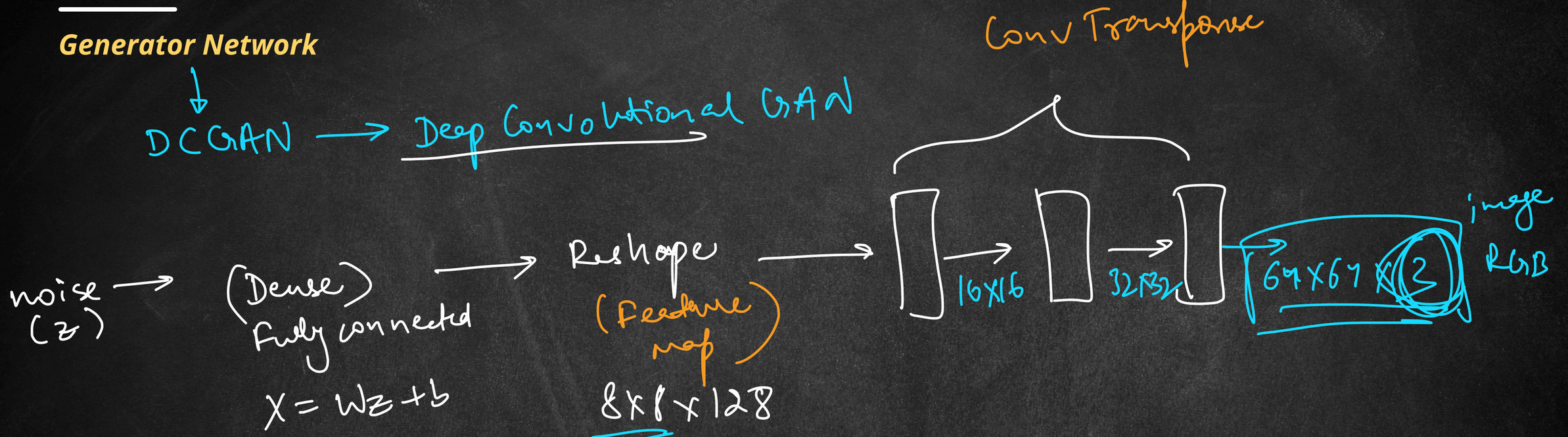
gaussian ($\mu = 0$, std dev = 1)

uniform

GANs

Generator Network

DCGAN → Deep Convolutional GAN



GANs



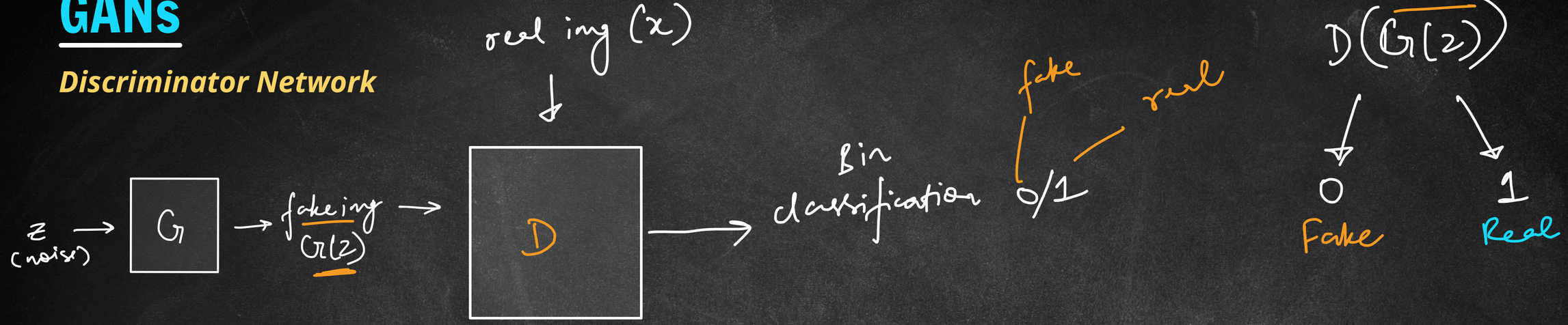
Generator Network - Code



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Discriminator Network



(1) Generator Loss (minimize)

$$g_loss = -\log(D(G(z)))$$

(1) $\rightarrow \infty$

(2) $\rightarrow 0$ ✓

(2) Discriminator Loss

$$d_loss = -[\log(D(x)) + \log(1 - D(G(z)))]$$

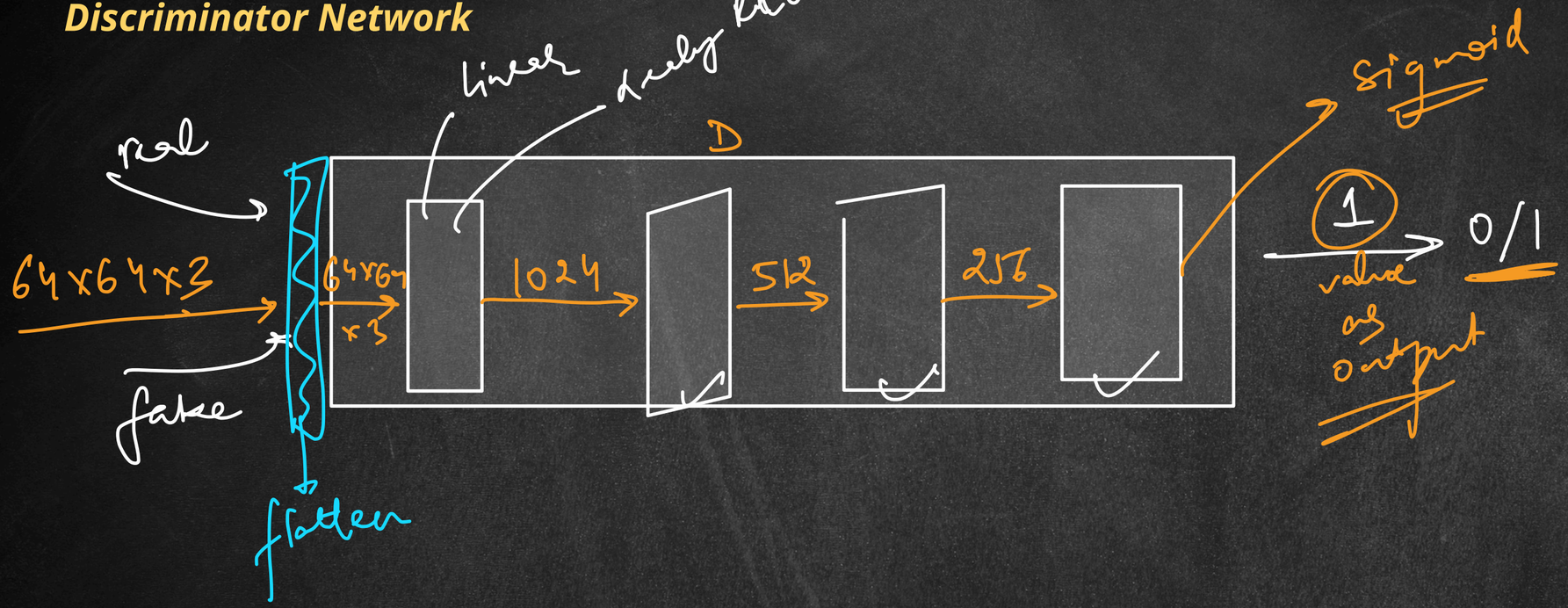
$$-[\log(1) + \log(1-0)] \rightarrow 0$$

GANs

→ vanilla GAN

CNN

Discriminator Network



Binary Cross Entropy

G → upsampling → generate

D → downsampling → classify

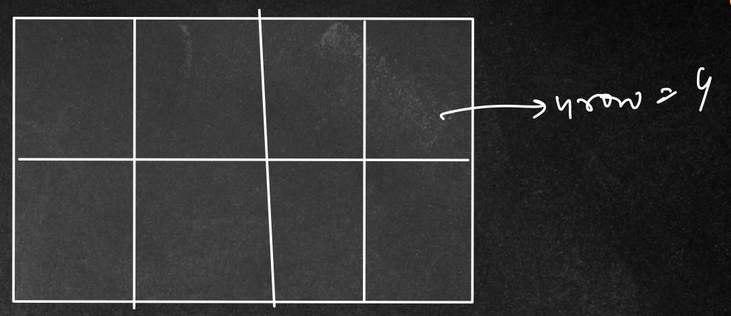
GANs

Discriminator Network - Code

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GANs

Training



grid tensor ⟹ channels x H x W
0 1 2

matplotlib ⟹ H x W x C
1 2 0